

ETHICAL HACKING-ITA1416

ASSESSMENT TOOL II – REVERSE ENGINEERING TASK (MEDIUM)

SLOT-B

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Task 1

1. Which functions or modules are responsible for gathering target information?

- WHOIS lookup module
- DNS lookup functions
- Network scanning module
- IP address enumeration functions
- HTTP/HTTPS request module
- OSINT data collection module

2. What footprinting techniques are implemented in the executable?

- WHOIS information gathering
- DNS enumeration
- IP address identification
- Domain and subdomain discovery
- Network reconnaissance
- Public information (OSINT) collection

3. How can the information collection process be disabled or modified?

- Remove or disable the footprinting functions.
- Block outbound network connections using a firewall.
- Modify the executable to bypass information-gathering routines.
- Restrict DNS and WHOIS queries.
- Replace network requests with dummy values during analysis.

Task 2

1. Which function is responsible for initiating DNS requests?

The DNS request function uses standard DNS APIs or socket functions to send DNS queries for records such as A, MX, NS, and SOA.

2. Does the application attempt an AXFR (Zone Transfer) request? Justify your answer.

Yes. If reverse engineering shows DNS queries requesting **AXFR**, it indicates the application attempts a DNS zone transfer to retrieve the complete DNS zone from the server.

3. What information can an attacker obtain using the extracted DNS records?

- Hostnames
- Subdomains
- IP addresses
- Mail servers (MX)
- Name servers (NS)
- Internal network structure
- DNS configuration details

Task 3

1. What types of information are collected by the application?

- Employee names
- Email addresses
- Organization details
- Domain names
- IP addresses
- Website technologies
- Social media profiles
- Public documents

2. Which external websites or services are accessed during execution?

- WHOIS databases
- DNS lookup services
- Search engines
- LinkedIn
- GitHub

- BuiltWith
- SimilarWeb
- Public company websites

3. How can the collected intelligence be misused by attackers?

- Spear-phishing attacks
- Social engineering
- Credential theft
- Reconnaissance for future attacks
- Password guessing
- Targeted malware campaigns

Task 4

1. Which functions are responsible for generating phishing emails?

- Email template generation module
- Message formatting functions
- SMTP email sending function
- Personalization module
- Attachment and link generation functions

2. What information is used to personalize the messages?

- Employee names
- Email addresses
- Organization name
- Department
- Job title
- Company branding
- Contact information

3. How can the phishing functionality be prevented or removed?

- Remove or disable email-generation modules.
- Block SMTP communication.

- Delete phishing templates.
- Prevent unauthorized execution using endpoint security.
- Detect and quarantine the application with antivirus software.

Task 5

1. How does the application capture usernames and passwords?

The application displays a fake login page that mimics a legitimate website. When users enter their credentials, the program captures the submitted username and password through the login form.

2. Where are the captured credentials stored or transmitted?

- Local files
- Databases
- Temporary storage
- Remote command-and-control (C2) server
- HTTP/HTTPS POST requests
- Email or cloud storage controlled by the attacker

3. What security risks arise from the use of such applications?

- Credential theft
- Unauthorized account access
- Identity theft
- Financial fraud
- Data breaches
- Privilege escalation
- Malware installation
- Organizational compromise